

Replication

ISE Coursework - Configuration Performance Learning Tool

1. Overview

This document provides complete, step-by-step instructions to replicate all experimental results reported in the coursework report Random Forest for Configuration Performance Learning. Following these instructions will reproduce: Table 3 (mean MAPE +/- std for representative datasets), Table 4 (aggregate W/T/L and significance counts), and Figures 1-3 as presented in the report.

2. Environment Requirements

Ensure the following are installed before running the tool:

Component	Version / Specification
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+ (any OS with Python 3.9+)
Python	3.9 or higher (3.11 recommended)
scikit-learn	$\geq 1.3.0$
pandas	$\geq 2.0.0$
numpy	$\geq 1.24.0$
scipy	$\geq 1.11.0$

3. Replication Steps

Step 1 - Clone the Repository

Clone the public repository to your local machine:

- `git clone https://github.com/cmpranav17/ise-cpl-coursework.git`
- `cd ise-cpl-coursework`

The repository contains all source code, the complete datasets/ folder (93 CSV files from the course Lab 2 data), and the three required PDF files in the root directory.

Step 2 - Install Dependencies

Install all required Python libraries in one command:

- `pip install scikit-learn pandas numpy scipy`

To verify successful installation, run:

- `python -c "import sklearn, pandas, numpy, scipy; print(sklearn.__version__, pandas.__version__, numpy.__version__, scipy.__version__)"`

Step 3 - Run the Main Evaluation Script

Execute Evaluate.py from the repository root. This runs all models on all 93 dataset-workload combinations and produces the raw results:

- `python Evaluate.py`

What the script does:

- Iterates over all 9 systems and all workloads under datasets/
- For each dataset, runs 30 independent 70/30 train-test splits (seeds 0-29)
- Trains LinearRegression, RandomForestRegressor (100 trees, max_features='sqrt'), and GaussianProcessRegressor (lrzip only) wrapped in StandardScaler Pipelines
- Computes MAPE, MAE, RMSE per split; applies the Wilcoxon signed-rank test across 30 repeats
- Saves results to results/all_results.csv and results/summary_table.csv

Expected runtime: 5-15 minutes. Progress is printed to stdout as each dataset finishes. No GPU is required.

Step 4 - Generate Figures

After Evaluate.py finishes, run the figure generation script:

- `python plot.py`

This produces three PNG files in results/ corresponding to Figures 1-3 of the report:

Output File	Report Figure	Description
results/figure1_mape_bar.png	Figure 1	Mean MAPE (log scale) bar chart for all 93 datasets - LR vs RF
results/figure2_pred_vs_actual.png	Figure 2	Predicted vs. actual scatter plots for 4 representative datasets
results/figure3_improvement.png	Figure 3	Per-dataset relative MAPE improvement (3a) and log LR vs RF

Step 5 - Verify Results Against the Report

Open results/all_results.csv and cross-check the mean MAPE values against Table 3 in the report.

Representative values to verify:

Dataset	LR Mean MAPE (%)	RF Mean MAPE (%)
batlik/village	56.77 +/- 2.40	4.22 +/- 0.33
dconvert/jpeg-large	13.57 +/- 0.27	2.95 +/- 0.06
kanzi/vmlinux	2855.03 +/- 137.57	350.52 +/- 39.03
xz/ambivert.wav.tar	135.85 +/- 6.86	15.18 +/- 2.03
z3/QF_BV_bench_935	1664.54 +/- 99.47	290.63 +/- 32.36
z3/QF_NRA_hong_9	0.21 +/- 0.01	2.55 +/- 0.18

Minor floating-point variations (< 0.5%) are acceptable due to platform-level random number generation differences. Values should match within the reported standard deviations.

Step 6 - Verify Aggregate Statistics (Table 4)

Open results/summary_table.csv and verify the aggregate W/T/L counts and significance numbers:

Model	Datasets	Better MAPE	Avg. Impr.	W	T	L	Sig. (p<0.05)
RandomForest vs LR	93	75/93	+35.2%	2326	0	464	85/93
GaussianProcess vs LR	13	1/13	-4.1%	125	0	265	9/13

4. Key Design Choices Affecting Reproducibility

The following design choices in Evaluate.py are critical to obtaining the reported results:

Choice	Value / Setting	Why It Matters
Random seeds	0 to 29 (30 seeds)	Controls train/test split randomness all models use the same splits per seed
Train/test ratio	70% / 30%	Fixed across all datasets and repeats
RF n_estimators	100	Default ensemble size - no tuning performed
RF max_features	'sqrt'	Standard regression forest setting sqrt of total feature count
Scaler	StandardScaler (in Pipeline)	) Applied to all models uniformly before fitting
GP threshold	< 500 rows	GP is only run on datasets with fewer than 500 configuration rows (lrzip) set;
Statistical test	Wilcoxon signed-rank (scipy)	Non-parametric paired test on the 30 MAPE values per data p < 0.05

5. Summary - Full Command Sequence

Copy and paste the following commands in order to perform a full replication:

- `git clone https://github.com/cmpranav17/ise-cpl-coursework.git`
- `cd ise-cpl-coursework`
- `pip install scikit-learn pandas numpy scipy`
- `python Evaluate.py`
- `python plot.py`

After completion, all results will be in results/ and all figures will be available as PNG files for direct comparison with the report.